Haptic-based Interfaces

Not unlike TUIs that we've just discussed, haptic interfaces rely on the user receiving physical feedback to their input actions. Simply put, tangible user interfaces require you to perform some sort of physical interaction with the system, whereas haptic interfaces will provide you with a sort of haptic feedback, be it in the form of a vibration, a compression—basically anything that you can physically feel.

Such interfaces can be used to enhance our gamut of senses by augmenting them in a way that we can perceive. This kind of technology is implemented in devices for physically challenged people such as those with visual impairments. An excellent example of a haptic interface in a commercially available device is "Lechal", smart footwear that guide you towards your destination using gentle vibrations.

Lechal utilizes metallic sensors (or pods) that are clipped onto the side of the shoe or placed in the insoles. The sensors are then paired to the user's smartphone via Bluetooth, who then uses the GPS-supported phone application that can be also via voice commands. Using real-time location tracking, the left or right pod will vibrate indicating the direction in which the user should turn.

Another cool example of haptic feedback is the Alert Shirt—a piece of apparel that aims to connect humans across vast distances by detecting the tangible, physical manifestations of human emotions and transmitting it for others to feel and share. The Alert Shirt was designed to let sports fans feel the physical strain and emotional stress that players feel when they are on the field. It does this using embedded haptic motors that respond to actions in the games, such as tackles, kicks, and even nerves that a player might feel when stepping up for a crucial kick. With the rise in popularity of wearables, it seems only evident that most devices in the future incorporate some sort of haptic feedback system.